

ARM-450 rev I

every joint driven, every part checked against every other

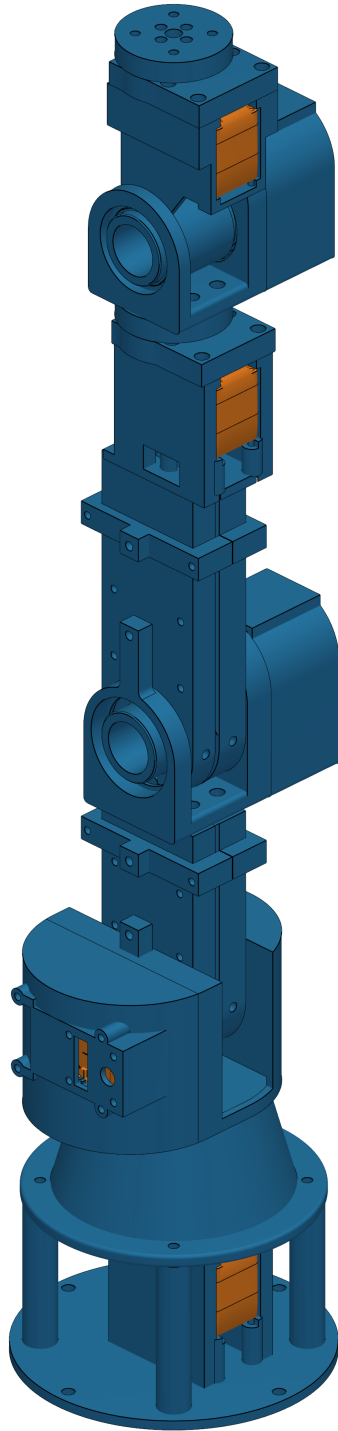
PRINT VERDICT (2026-09-24): GREEN -- 26 release stages at 0 failures.

- * all six joints driven by ST3215 servos through bolted horn couplers
- * every servo seated on 4 supports on its back and screwed by its own holes (20 screws)
- * every part checked against every other as exact solids, the manufacturer's servo model too
- * 265 dimensions measured back out of the print files; 114 screws checked as solids
- * assembly and disassembly paths free; screwdriver reaches every screw at its build step
- * cable route: one joint per cable, loops 31 .. 61 mm; nothing can wind up

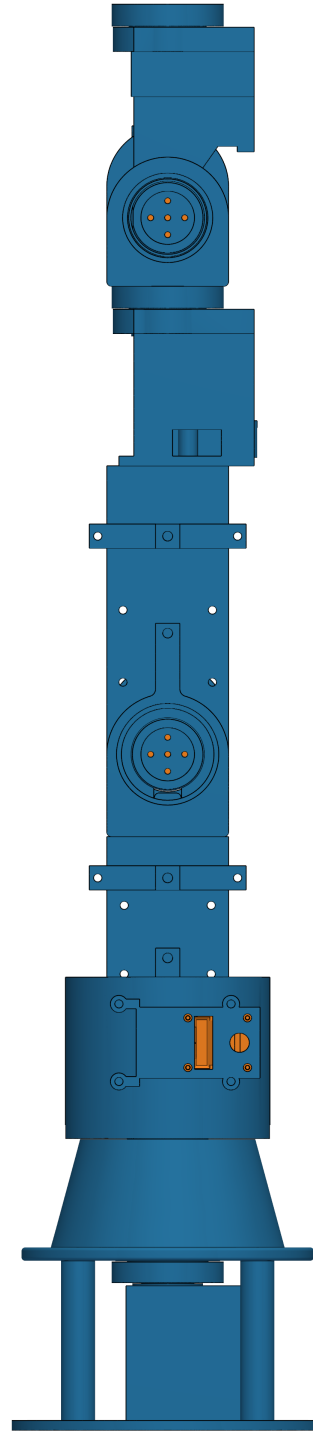
Read first: ARM450_AUDIT_REPORT.pdf. Details: ARM450_REV_I_FINAL.pdf, ARM450_POSES.pdf, ARM450_PARTS.pdf. Print: PRINTABLE_FILES (fit test first).

Full arm, 39 parts incl. 6 x ST3215

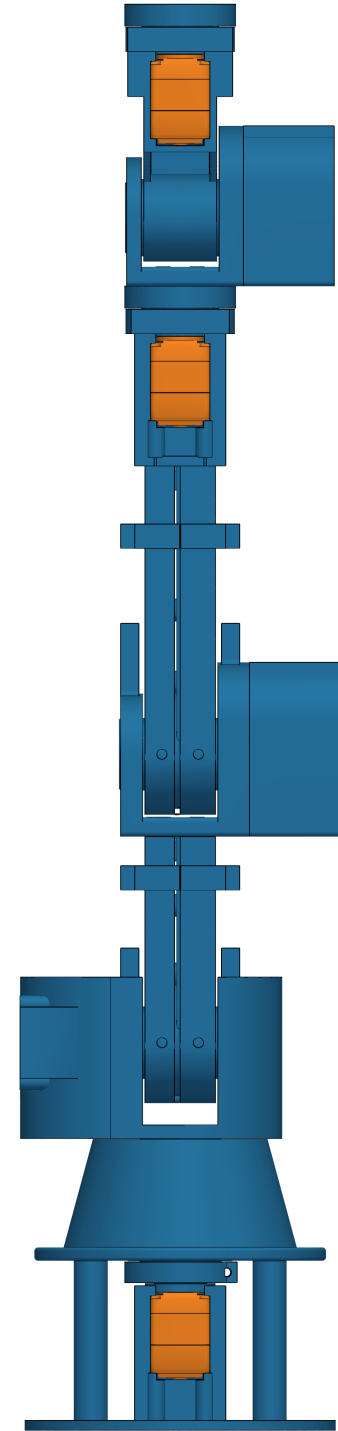
isometric



side (along the J2/J3/J5 axes)

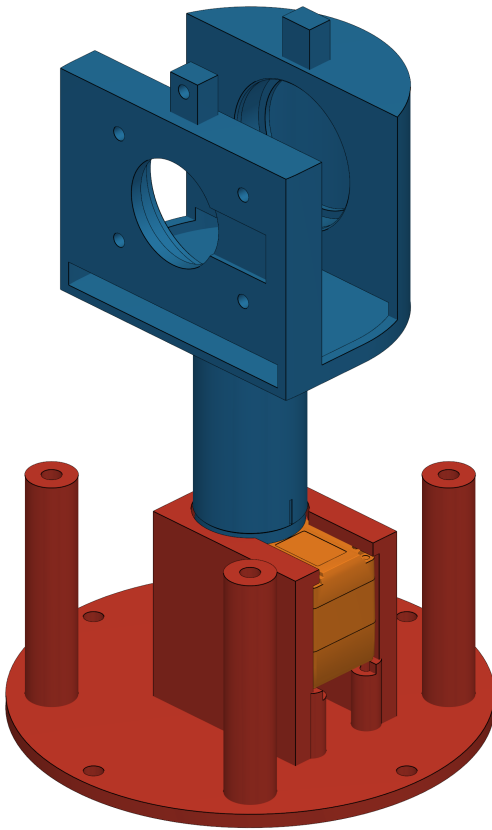


front

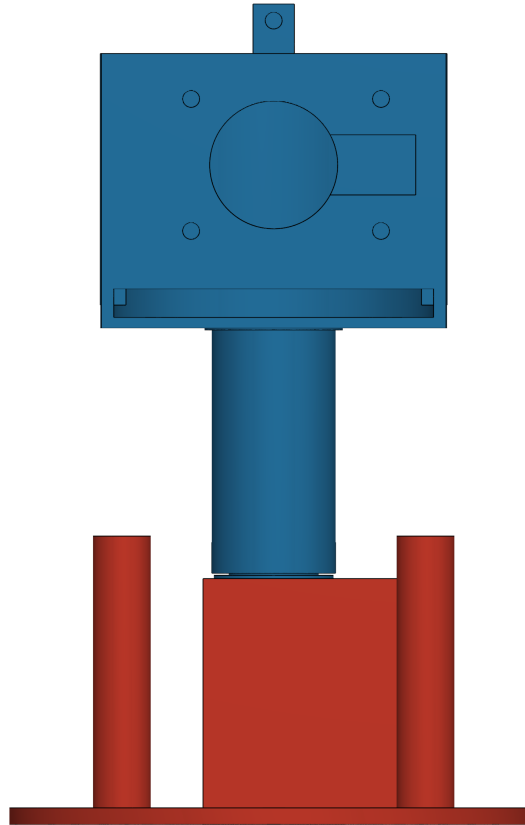


J1 base yaw

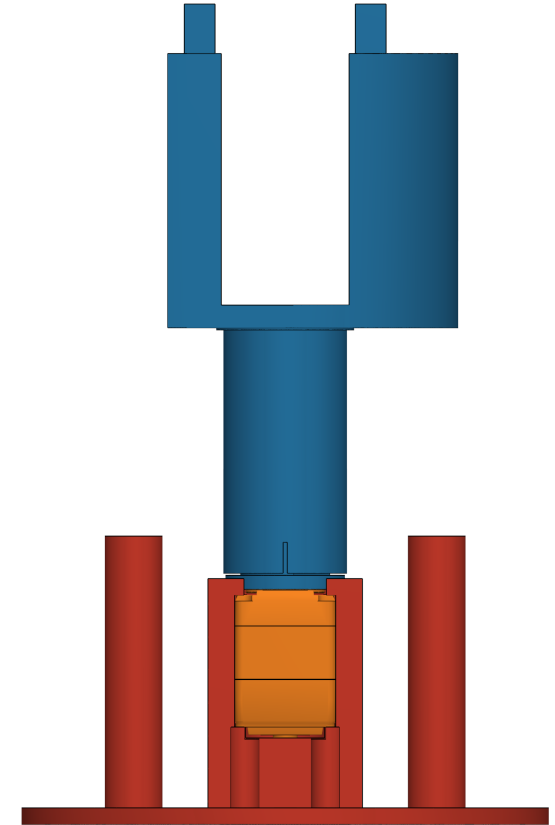
isometric



side (along the J2/J3/J5 axes)



front



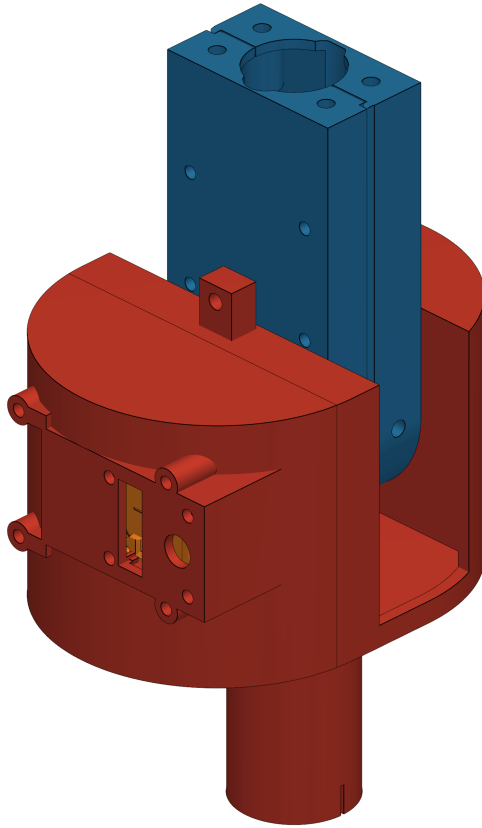
RED = fixed side (previous link): j1_mount
BLUE = driven (next link): j1_hub, J2_turret_p1
ORANGE = ST3215

Servo slides into the foot's pinch channel against the end wall; its front face under the lips. It stands on 4 posts on its back plateau and is screwed by its OWN 4 back holes from under the foot plate (the self-tappers from the servo box). Hub bolted to the horn (4 x M2.5); its D-plug keys the turret spigot, slit into a collet and clamped by the spigot collar. The turret runs in 2 x 6806 in the base; base on the foot's 4 columns (4 x M4); foot fixed to the table through its 4 holes.

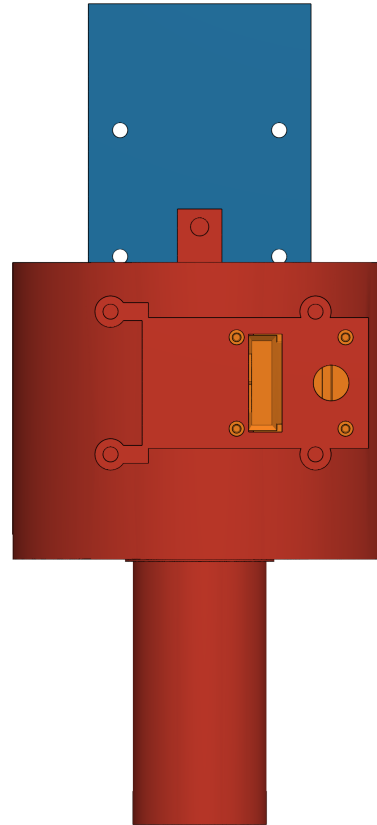
Verification: 0 failures (see REVI_CHECKS.log).

J2 shoulder

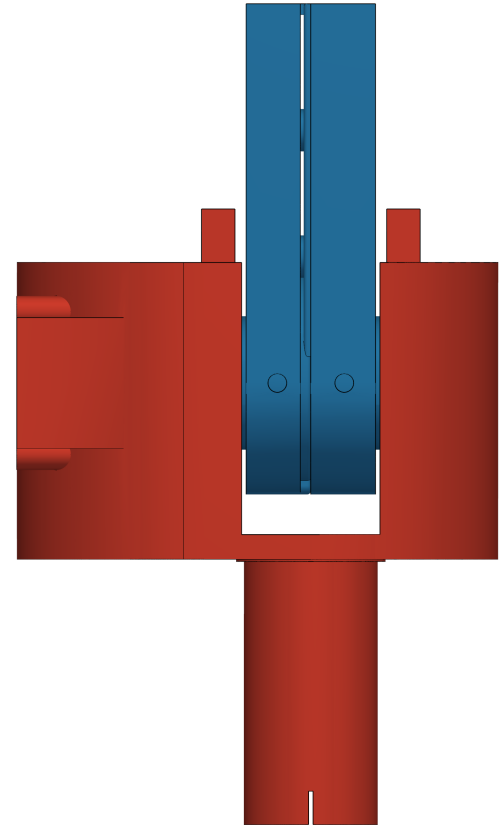
isometric



side (along the J2/J3/J5 axes)



front



RED = fixed side (previous link): J2_turret_p1, J2_turret_p2

BLUE = driven (next link): J2_shaft, link_upper_groove, link_upper_tongue, shaft_clamp_1, shaft_clamp_2

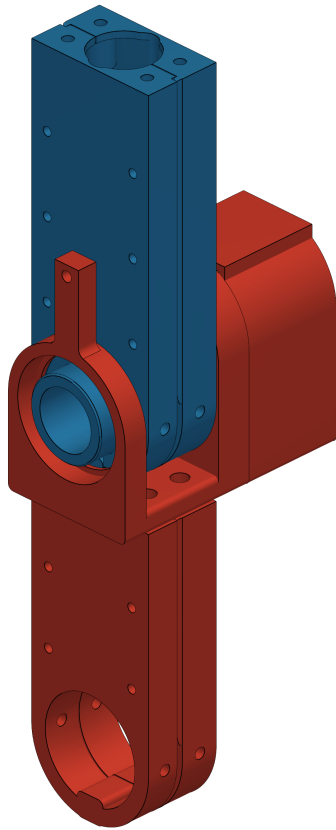
ORANGE = ST3215

Servo in the turret pod (-0.22 mm pinch), front face on the parting plane; the cover's 4 pads sit on its back plateau and 4 self-tappers through the cover go into its back holes. Cover: 4 x M3 x 45. Printed shaft bolted to the horn (4 x M2.5, driven through the shaft bore). Both 6806 press in from the fork gap. Link halves locked on the double-D shaft by 2 x M5 x 10 set screws each (centring rings).

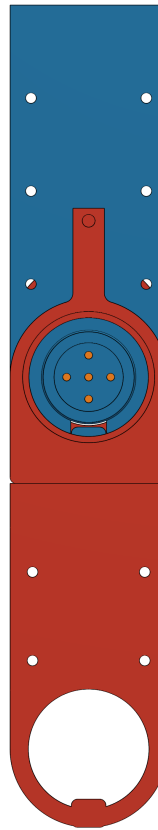
Verification: 0 failures (see REVI_CHECKS.log).

J3 elbow

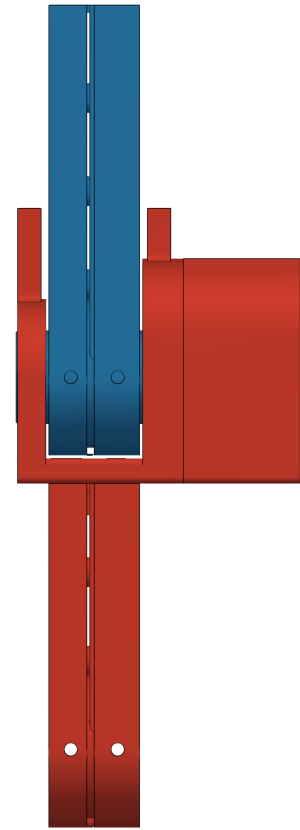
isometric



side (along the J2/J3/J5 axes)



front



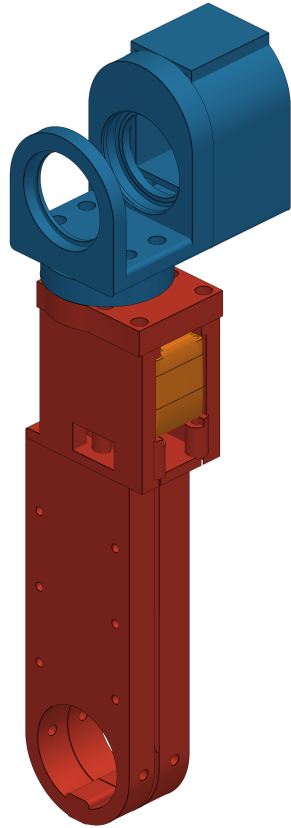
RED = fixed side (previous link): J3_p1, J3_p2, link_upper_groove, link_upper_tongue
BLUE = driven (next link): J3_shaft, link_fore_tongue, link_fore_groove, shaft_clamp_3, shaft_clamp_4
ORANGE = ST3215

Same drive train as J2. The fork bolts to the upper link's socket end with 4 x M3 x 10, heads sunk 3.2 mm in the fork floor under the swinging forearm. Servo on 4 cover pads + 4 servo screws.

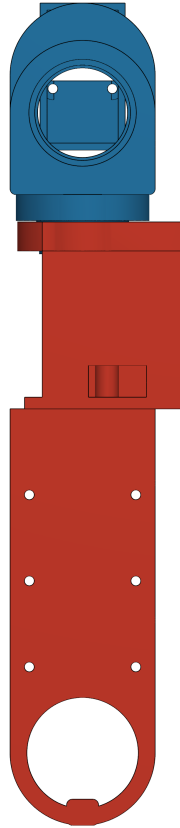
Verification: 0 failures (see REVI_CHECKS.log).

J4 forearm roll

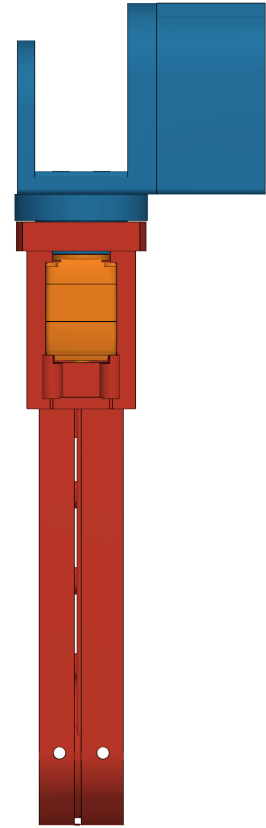
isometric



side (along the J2/J3/J5 axes)



front



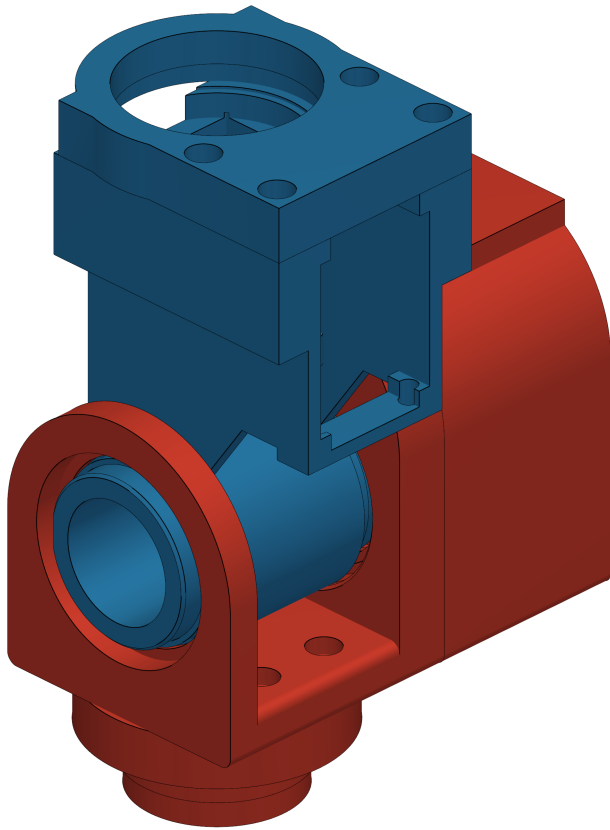
RED = fixed side (previous link): j4_base, j4_cap, link_fore_tongue, link_fore_groove
BLUE = driven (next link): j4_hub, J5_p1, J5_p2
ORANGE = ST3215

Base bolts to the forearm's socket end (4 x M3 + spigot, cables down the forearm). Servo slides in along its channel and stands on 4 posts (2 with servo screws from under the plate). Cap: 4 x M3 x 12, heads 3.8 deep, holds the 6706. Hub bolted to the horn; the J5 fork bolts to the hub's 4 inserts (M3 x 10, heads sunk in the fork floor).

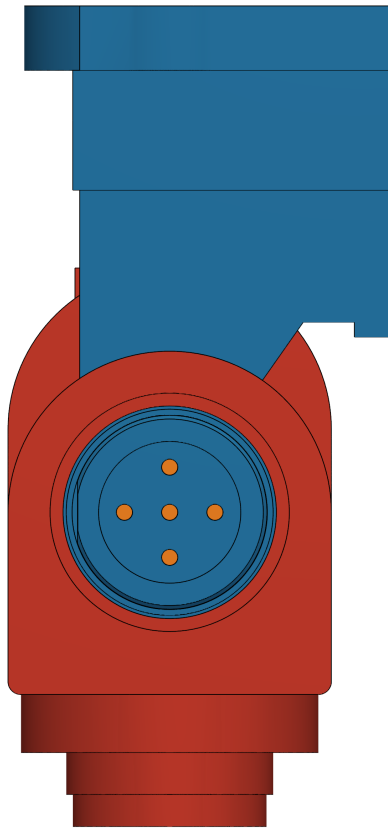
Verification: 0 failures (see REVI_CHECKS.log).

J5 wrist pitch

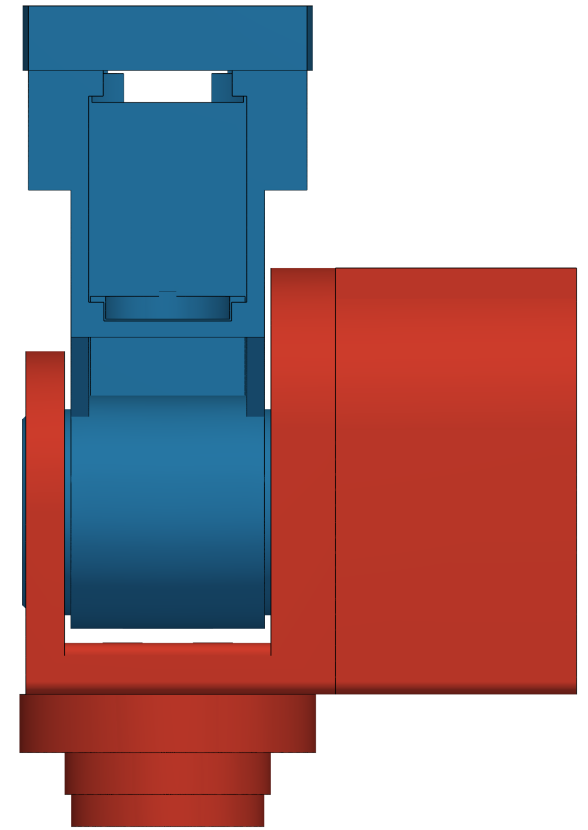
isometric



side (along the J2/J3/J5 axes)



front



RED = fixed side (previous link): J5_p1, J5_p2, j4_hub

BLUE = driven (next link): J5_shaft, J5_spacer_A, J5_spacer_B, j6_body, j6_cap

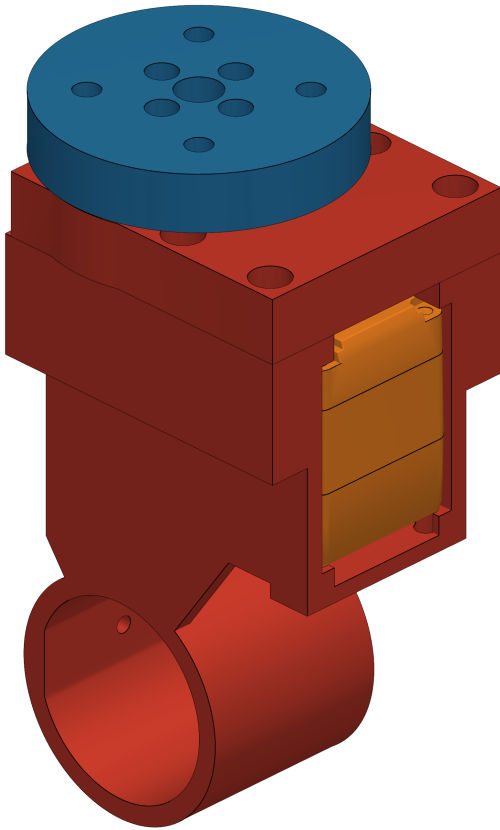
ORANGE = ST3215

Fork on the J3 rules; servo on 4 cover pads + 4 servo screws. Printed axle bolted to the horn; its flat runs out of the idle end into the blade's D-bore (+ M3 grub). The spacers clamp the blade between the inner races. Both 6706 press in from the gap.

Verification: 0 failures (see REVI_CHECKS.log).

J6 tool roll

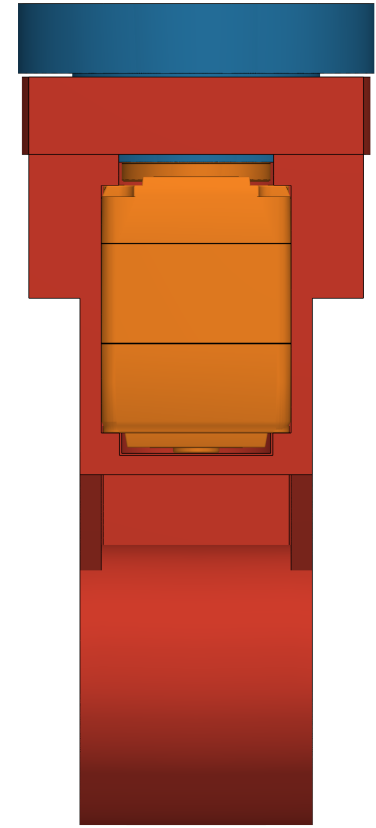
isometric



side (along the J2/J3/J5 axes)



front



RED = fixed side (previous link): j6_body, j6_cap
BLUE = driven (next link): j6_flange
ORANGE = ST3215

ST3215 in the blade, output along the arm; slides in along its channel and stands on 2 near pads + a far bridge with 2 servo screws; the cap holds a 6706. Tool flange bolted to the horn; 4 x M3 inserts on 030 for the tool.

Verification: 0 failures (see REVI_CHECKS.log).